LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Second laboratory (V2.1)

GENERAL INPUT/OUTPUT OF 32-BIT MICROCONTROLLERS. REGISTER LEVEL

Mandatory part (6 points):

- Prepare the necessary configuration of the project using the graphical interface of STM32CubeMX (**1 point**).
- Generate the code files of the project (**1 point**).
- Repeat the binary combinations of 0, 3, and 1 using the green and red LEDs. The values should change every 1 second (**2 points**).
- Write the program based on which the red LED should flash every one second and the green LED should flash every three seconds (**2 points**).

Additional parts (4 points):

- Make the green LED shine when the user button is pressed. The green LED should stop shining when the user button is released (**2 points**).
- Write the program to change the frequency of blinking of the red LED. Initially the red LED should blink every 1 second. The step of blinking should be changed using the user button. The period of blinking of the red LED should increase by 0.5 seconds after every press of the user button (**2 points**).

Note. All tasks must be performed without using the HAL library functions: *HAL_GPIO_Init()*, *HAL_GPIO_ReadPin()*, *HAL_GPIO_WritePin(), and HAL_GPIO_TogglePin()*. In other words, the content of all necessary registers should be set directly.

The ***DM00031020 STM32F4x reference manual registers the level of*** literature source provided in the Moodle system, which should be very valuable in this laboratory task.

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Second laboratory (V2.2)

GENERAL INPUT/OUTPUT OF 32-BIT MICROCONTROLLERS. REGISTER LEVEL

Mandatory part (6 points):

- Prepare the necessary configuration of the project using the graphical interface of STM32CubeMX (**1 point**).
- Generate the code files of the project (**1 point**).
- Repeat the binary combinations of 3, 2, and 1 using the green and red LEDs. The values should change every 1 second (**2 points**).
- Write the program based on which the red LED should flash every one second and the green LED should flash every two seconds (**2 points**).

Additional parts (4 points):

- Make the green LED shine when the user button is pressed. The green LED should remain shining when the user button is released. The red LED should stop shining when the user button is pressed repeatedly (**2 points**).
- Write the program to change the frequency of blinking of the green LED. Initially the green LED should blink every 1 second. The step of blinking should be changed using the user button. The period of blinking of the green LED should increase by 0.5 second after every press of the user button (**2 points**).

Note. All tasks must be performed without using the HAL library functions: *HAL_GPIO_Init()*, *HAL_GPIO_ReadPin()*, *HAL_GPIO_WritePin(), and HAL_GPIO_TogglePin()*. In other words, the content of all necessary registers should be set directly.

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LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Second laboratory (V2.3)

GENERAL INPUT/OUTPUT OF 32-BIT MICROCONTROLLERS. REGISTER LEVEL

Mandatory part (6 points):

- Prepare the necessary configuration of the project using the graphical interface of STM32CubeMX (**1 point**).
- Generate the code files of the project (**1 point**).
- Repeat the binary combinations of 0, 1, 2, and 3 using the green and red LEDs. The values should change every 1 second (**2 points**).
- Write the program based on which the red LED should flash every one second and the green LED should flash every two seconds (**2 points**).

Additional parts (4 points):

- Make the green LED shine when the user button is pressed. The green LED should remain shining when the user button is released. The green LED should stop shining when the user button is pressed repeatedly (**2 points**).
- Write the program to change the frequency of blinking of the red LED. Initially the red LED should blink every 1 second. The step of blinking should be changed using the user button. The period of blinking of the red LED should increase by 1 second after every press of the user button (**2 points**).

Note. All tasks must be performed without using the HAL library functions: *HAL_GPIO_Init()*, *HAL_GPIO_ReadPin()*, *HAL_GPIO_WritePin(), and HAL_GPIO_TogglePin()*. In other words, the content of all necessary registers should be set directly.

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LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Second laboratory (V2.4)

GENERAL INPUT/OUTPUT OF 32-BIT MICROCONTROLLERS. REGISTER LEVEL

Mandatory part (6 points):

- Prepare the necessary configuration of the project using the graphical interface of STM32CubeMX (**1 point**).
- Generate the code files of the project (**1 point**).
- Repeat the binary combinations of 0, 2, and 3 using the green and red LEDs. The values should change every 1 second (**2 points**).
- Write the program based on which the green LED should flash every one second and the red LED should flash every two seconds (**2 points**).

Additional parts (4 points):

- Make the green LED shine when the user button is pressed. The green LED should stop shining when the user button is released (**2 points**).
- Write the program to change the frequency of blinking of the red LED. Initially the red LED should blink every 1 second. The step of blinking should be changed using the user button. The period of blinking of the red LED should increase by 2 seconds after every press of the user button (**2 points**).

Note. All tasks must be performed without using the HAL library functions: *HAL_GPIO_Init()*, *HAL_GPIO_ReadPin()*, *HAL_GPIO_WritePin(), and HAL_GPIO_TogglePin()*. In other words, the content of all necessary registers should be set directly.

The ***DM00031020 STM32F4x reference manual registers the level of*** literature source provided in the Moodle system, which should be very valuable in this laboratory task.